# Classification Assignment

1) Problem Identification
    - Domain: **Machine Learning**.
    - It comes under **Supervised learning** because I/P & O/P are present.
    - It is **Classification** as the O/P is Categorical value.
    - We need to predict the Chronic Kidney Disease (CKD) based on the several parameters.
2) Basic info of dataset
    - Dataset has 399 rows and 25 columns.
3) Pre-Processing method
    - Dataset contains Columns "sg", "rbc", "pc", "pcc", "ba", "htn", "dm", "cad", "appet", "pe", "ane" and "classification". These data are Categorical data and also the nominal data.
    - It has been converted as numerical values (binary values (i.e) 1, 0).
    - I have used **one hot encoding** method to convert it as numerical data as this is Nominal data.
    - I don't have any weightage or rating data (i.e) ordinal data.
4) Model development
    - Models have been developed using Classification algorithms and uploaded the ipython files along with this document.
5) I have used GridSearchCV to play with multiple hypertuning parameters for each algorithm. I have uploaded the Ranking Score tables for each algorithm along with this document.

6) Evaluation metrics and best parameters for each algorithm are given below.

A. Support Vector Machine Classification

```
Evaluation metrics details for SVM are given below
1) Confusion Matrix:
 [[44  1]
 [ 0 75]]
2) classification report:
               precision    recall  f1-score   support

           0       1.00      0.98      0.99        45
           1       0.99      1.00      0.99        75

    accuracy                           0.99       120
   macro avg       0.99      0.99      0.99       120
weighted avg       0.99      0.99      0.99       120

3) roc_auc_score is 0.9988148148148148
4) The best parameteers from GridSearchCV for the created SVM model are :
 {'C': 0.1, 'kernel': 'linear', 'probability': True}
```

# Classification Assignment

## B. Decision Tree Classification

```
Evaluation metrics details for DT are given below
1) Confusion Matrix:
 [[45  0]
 [ 7 68]]
2) classification report:
               precision    recall  f1-score   support

           0       0.87      1.00      0.93        45
           1       1.00      0.91      0.95        75

    accuracy                           0.94       120
   macro avg       0.93      0.95      0.94       120
weighted avg       0.95      0.94      0.94       120

3) roc_auc_score is 0.9533333333333334
4) The best parameters from GridSearchCV for the created DT model are :
 {'criterion': 'entropy', 'max_features': 'sqrt', 'splitter': 'random'}
```

## C. Random Forest Classification

```
Evaluation metrics details for RF are given below
1) Confusion Matrix:
 [[45  0]
 [ 1 74]]
2) classification report:
               precision    recall  f1-score   support

           0       0.98      1.00      0.99        45
           1       1.00      0.99      0.99        75

    accuracy                           0.99       120
   macro avg       0.99      0.99      0.99       120
weighted avg       0.99      0.99      0.99       120

3) roc_auc_score is 1.0
4) The best parameters from GridSearchCV for the created RF model are :
 {'criterion': 'gini', 'max_features': 1, 'n_estimators': 100}
```

## D. Logistic Regression Classification

```
Evaluation metrics details for Logistic Regression Classification are given below
1) Confusion Matrix:
 [[43  2]
 [ 0 75]]
2) classification report:
               precision    recall  f1-score   support

           0       1.00      0.96      0.98        45
           1       0.97      1.00      0.99        75

    accuracy                           0.98       120
   macro avg       0.99      0.98      0.98       120
weighted avg       0.98      0.98      0.98       120

3) roc_auc_score is 0.9976296296296296
4) The best parameters from GridSearchCV for the created Logistic Regression Classification model are :
 {'C': 1.0, 'solver': 'liblinear'}
```

# Classification Assignment

E. Bernoulli Naive Bayes

```
Evaluation metrics details for BernoulliNB are given below
1) Confusion Matrix:
 [[45  0]
 [ 8 67]]
2) classification report:
               precision    recall  f1-score   support

           0       0.85      1.00      0.92        45
           1       1.00      0.89      0.94        75

    accuracy                           0.93       120
   macro avg       0.92      0.95      0.93       120
weighted avg       0.94      0.93      0.93       120

3) roc_auc_score is 0.9967407407407407
4) The best parameters from GridSearchCV for the created BernoulliNB model are :
 {'alpha': 1.0, 'binarize': 0.0}
```

F. Complement Naïve Bayes

```
Evaluation metrics details for ComplementNB are given below
1) Confusion Matrix:
 [[44  1]
 [22 53]]
2) classification report:
               precision    recall  f1-score   support

           0       0.67      0.98      0.79        45
           1       0.98      0.71      0.82        75

    accuracy                           0.81       120
   macro avg       0.82      0.84      0.81       120
weighted avg       0.86      0.81      0.81       120

3) roc_auc_score is 0.9099259259259259
4) The best parameters from GridSearchCV for the created ComplementNB model are :
 {'alpha': 1.0, 'norm': False}
```

G. Gaussian Naïve Bayes

```
Evaluation metrics details for GaussianNB are given below
1) Confusion Matrix:
 [[45  0]
 [ 2 73]]
2) classification report:
               precision    recall  f1-score   support

           0       0.96      1.00      0.98        45
           1       1.00      0.97      0.99        75

    accuracy                           0.98       120
   macro avg       0.98      0.99      0.98       120
weighted avg       0.98      0.98      0.98       120

3) roc_auc_score is 1.0
4) The best parameters from GridSearchCV for the created GaussianNB model are :
 {'var_smoothing': 1e-09}
```

H. Multinomial Naïve Bayes

```
Evaluation metrics details for MultinomialNB are given below
1) Confusion Matrix:
 [[44  1]
 [20 55]]
2) classification report:
               precision    recall  f1-score   support

           0       0.69      0.98      0.81        45
           1       0.98      0.73      0.84        75

    accuracy                           0.82       120
   macro avg       0.83      0.86      0.82       120
weighted avg       0.87      0.82      0.83       120

3) roc_auc_score is 0.9315555555555557
4) The best parameters from GridSearchCV for the created MultinomialNB model are :
 {'alpha': 0.1}
```

**7. Final Model**

In RF Confusion Matrix, **Type 1 error is 0** & **Type 2 error is 1** that is lesser than Gaussian NB.

**The best roc_auc_score value is 1**. It is found in **Random Forest & Gaussian NB**.

However, the other values such **precision, recall, f1_score, accuracy, macro average & Weighted Average** are **really high** in the model created by Random Forest.

```
Evaluation metrics details for RF are given below
1) Confusion Matrix:
 [[45  0]
 [ 1 74]]
2) classification report:
               precision    recall  f1-score   support

           0       0.98      1.00      0.99        45
           1       1.00      0.99      0.99        75

    accuracy                           0.99       120
   macro avg       0.99      0.99      0.99       120
weighted avg       0.99      0.99      0.99       120

3) roc_auc_score is 1.0
4) The best parameters from GridSearchCV for the created RF model are :
 {'criterion': 'gini', 'max_features': 1, 'n_estimators': 100}
```

```
Evaluation metrics details for GaussianNB are given below
1) Confusion Matrix:
 [[45  0]
 [ 2 73]]
2) classification report:
               precision    recall  f1-score   support

           0       0.96      1.00      0.98        45
           1       1.00      0.97      0.99        75

    accuracy                           0.98       120
   macro avg       0.98      0.99      0.98       120
weighted avg       0.98      0.98      0.98       120

3) roc_auc_score is 1.0
4) The best parameters from GridSearchCV for the created GaussianNB model are :
 {'var_smoothing': 1e-09}
```